BEING UNRAMIFIED, RESPECTIVELY G-UNRAMIFIED, IS FPQC LOCAL ON THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author's closing phrase 'Hence the result follows formally.' is quoted verbatim.

Editorial note. The counted spans carry navigation stubs for the references leaving them; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Characterisation of unramified and G-unramified morphisms by the diagonal being an open immersion. Morphisms, Lemma morphisms-lemma-diagonal-unramified-morphism, invoked at proof lines 6412-6415. Splits each property into a finiteness condition and openness of the diagonal.
- Local finite type, local finite presentation and open immersions are fpqc local on the base. Same chapter, Lemmas lemma-descending-property-locally-finite-presentation, lemma-descending-property-locally-finite-type and lemma-descending-property-open-immersion, invoked at proof lines 6416-6421. Gives fpqc locality of each ingredient.

Source reference. Original lemma 02KY: sources/stacks-descent/pinned-original-source.tex, source lines 5984–5984. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02KX: sources/stacks-descent/pinned-original-source.tex, source lines 5954–5954. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02L3: sources/stacks-descent/pinned-original-source.tex, source lines 6079–6079. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Source reference. Original lemma 02GE: sources/stacks-chow/tags.txt, source lines 3225–3225. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *The property $\mathcal{P}(f) =$"f is unramified" is fpqc local on the base. The property $\mathcal{P}(f) =$"f is G-unramified" is fpqc local on the base.*

Proof. A morphism is unramified (resp. G-unramified) if and only if it is locally of finite type (resp. finite presentation) and its diagonal morphism is an open immersion (see Morphisms, Lemma 02GE). We have seen already that being locally of finite type (resp. locally of finite presentation) and an open immersion is fpqc local on the base (Lemmas 02KY, 02KX, and 02L3). Hence the result follows formally. □